

Assignment 2

Due April 30 at Noon

Part A: Ethnic disparities in mathematics achievement growth and the contribution of school poverty concentration.

Your first task is to compare the average growth rates in math achievement of three ethnic groups in the Chicago elementary school data (EG1.sav, EG2.sav, EG3.sav). The three groups are Black students, Hispanic students, and the reference group which we refer to as white students (this group is overwhelming white in this sample). Construct a 3-level quadratic growth model for math achievement. The Year variable is centered at approximately the spring of 1st grade. For simplicity, do not center the indicators for “Black” and Hispanic”. Allow the child-level terms to vary. At the school level, only allow the terms for the reference group to vary (u_{00j} , u_{10j} , u_{20j}).

1. Write down the model at each level, and define terms in the model.
2. Run the model and interpret the output for the fixed coefficients (the gammas), explaining the findings with respect to ethnic differences in learning rates.
3. Create a model graph of the average trajectories and describe what you see in the graph. (Note: In this data, Black and Hispanic are mutually exclusive categories, so you can ignore a model-based prediction for students that are both).
4. You allowed only three terms to vary in your level-3 model. Explain what this means for your analysis. What are you assuming?
5. Construct and interpret 95% plausible value intervals for the child-level variation in achievement for white students, ~~the child-level Hispanic-white gap in average achievement~~, and the school-level variation in achievement for white students.

Your second task is to re-estimate these results, controlling for the school's poverty concentration (operationalized as the percent of low-income students in the school, LOWINC).

6. Do you find evidence that school poverty concentration predicts a child's growth rate? Present your revised model and its output. You not need to interpret every coefficient, only relevant ones.
7. Compare these results to those in question 2 above. Comment briefly: have the results changed? If so, how?

Part B: Revising sector effects

In assignment one, we considered the effects of various school- and student-level covariates on a continuous outcome. Now, we consider a dichotomous outcome. We will use data from the National Education Longitudinal Study. Despite the name of the dataset, we are using only a single cross-section of data, students nested within schools. The data have already been packaged in an .mdm file on Canvas called *nels_hm.mdm*. The outcome variable is BAGRAD, indicating that the student eventually obtained a BA degree.

1. First, run a model predicting college graduation with gender, race (note there are ~~four~~ three non-white racial categories), and base year SES (BYSES). Enter all covariates uncentered, and only allow the intercept to vary. Remember to use EM LaPlace estimation when running logistic regression models in HLM (1000 max iterations). Explain the sampling model and link function. Interpret the reported unit-specific estimates for the intercept and coefficients for FEMALE, BLACK, HISPANIC, ASIAN, and BYSES.
2. Calculate the model-based predicted probability of college graduation for the following people. Assume that other variables all equal 0 and use the unit-specific estimates for your predictions. Show the basis for your calculations.
 - a. a white male
 - b. a white female
 - c. a Hispanic male
 - d. a Hispanic female
3. Explain why the change in probability associated with being female is different for Hispanic and white students.
4. We are interested in whether Catholic schools promote higher college graduation rates in the NELS sample. Investigate this question by introducing the variables for sector and appropriate control variables at level 2. Interpret the relevant model coefficients for the population-average estimates.

Potential control variables are: URBAN (coded 1 if in an urban area, 0 otherwise), PCTMIN_S (percent minority students), SCHSIZE (school size), and FRELUNCH (percent of students that qualify for free lunch in schools; a rough measure of family poverty).

5. Give a summary about what you've learned about college graduation rates in the sample in substantive terms. Include an explanation of the "effect" of attending a Catholic high school, referencing an appropriate odds ratio *and* a change in probability associated with attending a Catholic school. Hint: for the latter, you should also explain who the change in probability is estimated for.